Two Factor Designs

~ Adventures in Model Building ~

RECALL

Types of Effects

Inference

RBI MBI

1) Contrasts $\widehat{ATE}_{1,2} = \hat{\bar{Y}}_1 - \hat{\bar{Y}}_2$

2) Effect Relative to Benchmark $\hat{\alpha}_j = \hat{\bar{Y}}_j - \hat{\bar{\bar{Y}}}$

3) F-Statistic $$F = \frac{MS_B}{MS_W} = \frac{\frac{1}{J-1}\sum_{j=1}^{J} n_j \hat{\alpha}_j^2}{\frac{1}{n-J}\sum_{i=1}^{n}\left(Y_i - \hat{\bar{Y}}_{j(i)}\right)^2}$$

MBI: 3 Models for Y_i

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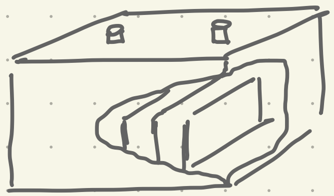
[Slides]

STUDY

Battery Life



Researchers wish to understand the effect that temperature (15, 70, 125) and material (1, 2, 3) have on the lifetime of a battery. They have access to 36 batteries.

Q Which design and analyis would you favor and why?

DEF

Two-way Factorial Design

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